

Configuring Apache2 Server and Implementing Firewall rules for IP Blocking

Goal:

- ✧ Set up networking between Windows PC and Kali Linux VM, install and configure Apache2 on Kali, host a test HTML page, access it from Windows using the Kali IP address, identify the Windows IP address from Apache logs, and finally block that Windows IP using the firewall on Kali Linux.

Objectives:

- Establish network communication between Windows and Kali.
- Install and configure Apache2 web server in Kali Linux.
- Host a custom HTML test page on the server.
- Access the web page from the Windows system using Kali's IP.
- Identify the Windows client IP from Apache logs.
- Block the Windows IP using firewall rules.

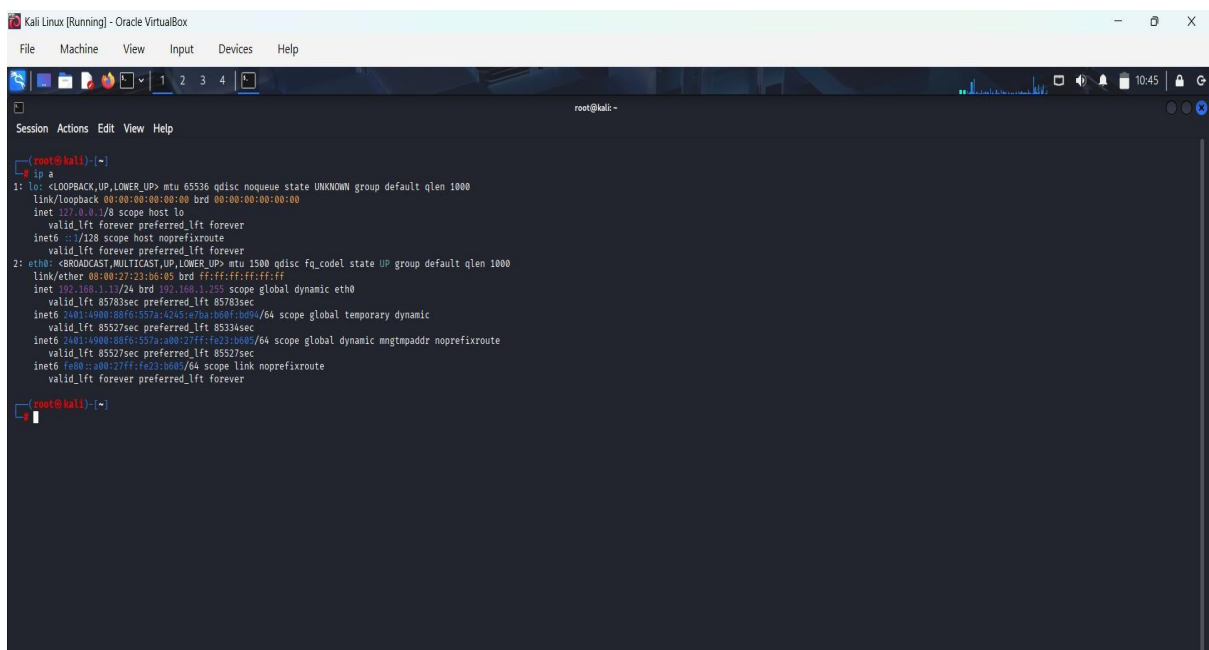
Step-by-Step Procedure

Step 1: Check and Configure Network Connection

1. Open VirtualBox / VMware settings for your Kali Linux VM.
2. Under Network, select Bridged Adapter (recommended) so Kali gets an IP from router.
3. Start Kali Linux.

Verify ip address:

ip a



```
root@kali:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:23:b6:05 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.9/24 brd 192.168.1.255 scope global dynamic eth0
        valid_lft 85783sec preferred_lft 85783sec
    inet6 2401:4900:b9f6:557a:a00:27ff:fe23:b605/64 scope global temporary dynamic
        valid_lft 85527sec preferred_lft 85334sec
    inet6 2401:4900:b9f6:557a:a00:27ff:fe23:b605/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 85527sec preferred_lft 85527sec
    inet6 fe80::a00:27ff:fe23:b605/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Step 2: Check IP Address in Windows Computer

ipconfig

Note down the ipv4 Address :192.168.1.9

Step 3: Verify Communication Between Kali and Windows

OnKali:

```
ping -c 4 : 192.168.1.9
```

On Windows:

```
Ping 192.168.1.13
```

If ping works → Both systems are connected correctly.

Step 4: Install Apache Web Server on Kali

```
sudo apt update  
sudo apt install apache2 -y
```

Start and enable Apache:

```
sudo systemctl enable apache2  
sudo systemctl status apache2
```

Active:active (running)

```
(root@kali) ~#  
-# sudo apt install -y apache2  
apache2 is already the newest version (2.4.65-3+b1).  
Summary:  
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 859  
  
(root@kali) ~#  
-# sudo systemctl start apache2  
  
(root@kali) ~#  
-# sudo systemctl enable apache2  
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.  
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2  
  
(root@kali) ~#  
-# sudo systemctl status apache2  
● apache2.service - The Apache HTTP Server  
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: disabled)  
   Active: active (running) since Wed 2025-11-05 10:34:38 IST; 30min ago  
     Invocation: d73e722e0a4133ab9e5b9709e278cb  
       Docs: https://httpd.apache.org/docs/2.4/  
    Main PID: 796 (apache2)  
      Tasks: 6 (limit: 4546)  
    Memory: 31.5M (peak: 31.9M)  
       CPU: 67ms  
    CGroup: /system.slice/apache2.service  
            └─796 /usr/sbin/apache2 -k start  
              └─799 /usr/sbin/apache2 -k start  
                └─800 /usr/sbin/apache2 -k start  
                  └─801 /usr/sbin/apache2 -k start  
                    └─802 /usr/sbin/apache2 -k start  
                      └─803 /usr/sbin/apache2 -k start  
  
Nov 05 10:34:37 kali systemd[1]: Starting apache2.service - The Apache HTTP Server...  
Nov 05 10:34:38 kali apache2ctl[789]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress this message  
Nov 05 10:34:38 kali systemd[1]: Started apache2.service - The Apache HTTP Server.
```

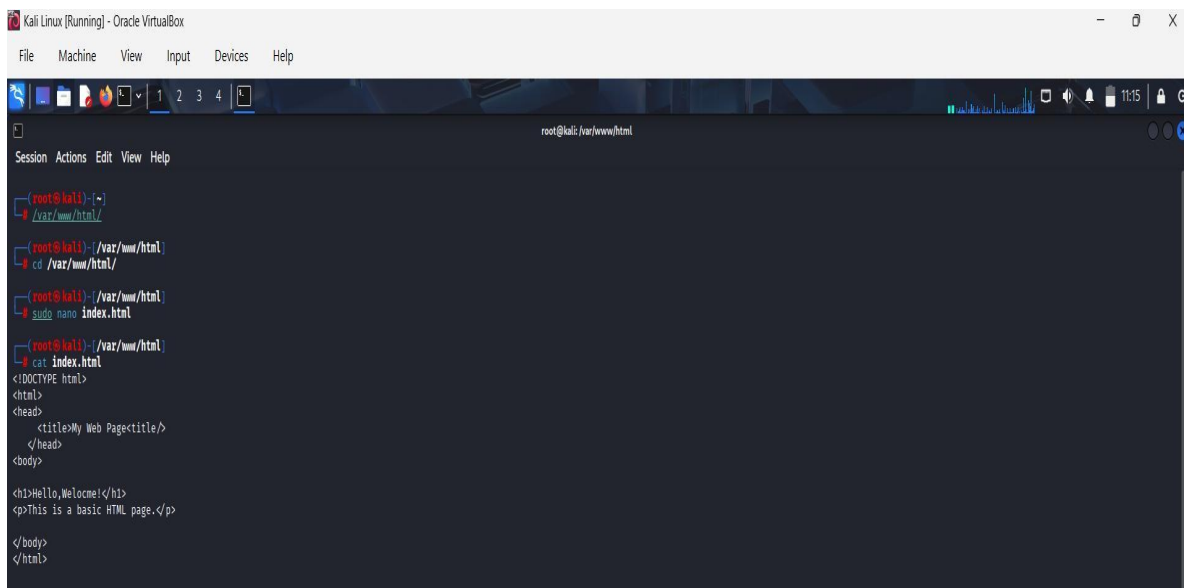
Step 5: Create a Simple HTML Web Page

```
/var/www/html  
cd /var/www/html  
sudo nano index.html
```

Add below HTML code:

```
<!DOCTYPE html>  
<html>  
<head>  
    <title>My Web Page</title>  
</head>  
<body>  
  
    <h1>Hello,Welcome!</h1>  
    <p>This is a basic HTML page.</p>  
  
</body>  
</html>
```

Save using: Ctrl + O → Enter → Ctrl + X



The screenshot shows a terminal window titled 'Kali Linux [Running] - Oracle VM VirtualBox'. The terminal output shows the following commands and their results:

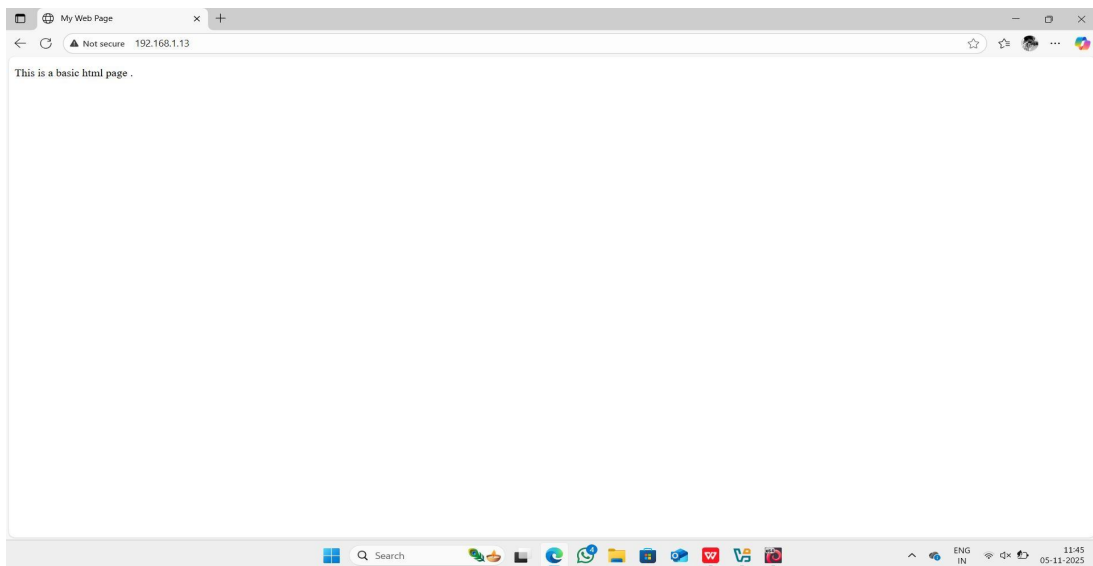
```
root@kali:~# cd /var/www/html/  
root@kali:~/var/www/html# cd /var/www/html/  
root@kali:~/var/www/html# sudo nano index.html  
root@kali:~/var/www/html# cat index.html  
<!DOCTYPE html>  
<html>  
<head>  
    <title>My Web Page</title>  
</head>  
<body>  
  
    <h1>Hello,Welcome!</h1>  
    <p>This is a basic HTML page.</p>  
  
</body>  
</html>
```

Step 6: Access the Web Page from Windows

Open browser on Windows Computer

Type: `http:// 192.168.1.13`

You should see your webpage as shown below:



Step 7: Check Access Logs to Identify Windows IP

`/var/log/apache2`

`sudo tail -n 20 /var/log/apache2/access.log`

You will see the logs as show in the below figure:

```
Kali Linux [Running] --Oracle VirtualBox
File Machine View Input Devices Help

root@kali: /var/log/apache2
--(root@kali) /var/log/apache2
--(root@kali) /var/log/apache2
-- sudo tail -f /var/log/apache2/access.log
192.168.1.9 - - [05/Nov/2025:11:32:10 +0530] "-" 408 0 "-" "-"
192.168.1.9 - - [05/Nov/2025:11:32:10 +0530] "-" 408 0 "-" "-"
192.168.1.9 - - [05/Nov/2025:11:32:40 +0530] "-" 408 0 "-" "-"
192.168.1.12 - - [05/Nov/2025:11:40:57 +0530] "GET / HTTP/1.1" 200 488 "android-app://com.google.android.googlequicksearchbox/" "Mozilla/5.0 (Linux; Android 10; K) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Mobile Safari/537.36"
192.168.1.9 - - [05/Nov/2025:11:41:10 +0530] "GET / HTTP/1.1" 200 488 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36 Edg/142.0.0.0"
192.168.1.9 - - [05/Nov/2025:11:41:50 +0530] "-" 408 0 "-" "-"
192.168.1.9 - - [05/Nov/2025:11:43:31 +0530] "GET / HTTP/1.1" 200 462 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36 Edg/142.0.0.0"
192.168.1.9 - - [05/Nov/2025:11:43:23 +0530] "-" 408 0 "-" "-"
192.168.1.9 - - [05/Nov/2025:11:44:59 +0530] "GET / HTTP/1.1" 200 469 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36 Edg/142.0.0.0"
192.168.1.9 - - [05/Nov/2025:11:45:50 +0530] "-" 408 0 "-" "-"
192.168.1.9 - - [05/Nov/2025:11:50:34 +0530] "GET / HTTP/1.1" 200 469 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36 Edg/142.0.0.0"
192.168.1.9 - - [05/Nov/2025:11:50:35 +0530] "GET / HTTP/1.1" 200 468 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36 Edg/142.0.0.0"
192.168.1.12 - - [05/Nov/2025:11:50:49 +0530] "GET / HTTP/1.1" 200 469 "android-app://com.google.android.googlequicksearchbox/" "Mozilla/5.0 (Linux; Android 10; K) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Mobile Safari/537.36"
192.168.1.12 - - [05/Nov/2025:11:50:52 +0530] "GET / HTTP/1.1" 200 468 "android-app://com.google.android.googlequicksearchbox/" "Mozilla/5.0 (Linux; Android 10; K) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Mobile Safari/537.36"
192.168.1.9 - - [05/Nov/2025:11:51:26 +0530] "-" 408 0 "-" "-"
192.168.1.9 - - [05/Nov/2025:11:51:35 +0530] "GET / HTTP/1.1" 200 469 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36 Edg/142.0.0.0"
192.168.1.9 - - [05/Nov/2025:11:51:36 +0530] "GET / HTTP/1.1" 200 468 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36 Edg/142.0.0.0"
192.168.1.9 - - [05/Nov/2025:11:51:36 +0530] "GET / HTTP/1.1" 200 468 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36 Edg/142.0.0.0"
192.168.1.12 - - [05/Nov/2025:11:52:01 +0530] "GET / HTTP/1.1" 200 469 "android-app://com.google.android.googlequicksearchbox/" "Mozilla/5.0 (Linux; Android 10; K) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Mobile Safari/537.36"
192.168.1.12 - - [05/Nov/2025:11:52:04 +0530] "GET / HTTP/1.1" 200 468 "android-app://com.google.android.googlequicksearchbox/" "Mozilla/5.0 (Linux; Android 10; K) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Mobile Safari/537.36"
192.168.1.12 - - [05/Nov/2025:11:52:05 +0530] "GET / HTTP/1.1" 200 468 "android-app://com.google.android.googlequicksearchbox/" "Mozilla/5.0 (Linux; Android 10; K) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Mobile Safari/537.36"
192.168.1.9 - - [05/Nov/2025:11:52:16 +0530] "GET / HTTP/1.1" 200 469 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36 Edg/142.0.0.0"
192.168.1.9 - - [05/Nov/2025:11:52:15 +0530] "GET / HTTP/1.1" 200 468 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36 Edg/142.0.0.0"
192.168.1.9 - - [05/Nov/2025:11:53:00 +0530] "-" 408 0 "-" "-"
192.168.1.9 - - [05/Nov/2025:11:53:49 +0530] "GET / HTTP/1.1" 200 469 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36 Edg/142.0.0.0"
192.168.1.9 - - [05/Nov/2025:11:53:43 +0530] "GET / HTTP/1.1" 200 468 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36 Edg/142.0.0.0"
192.168.1.9 - - [05/Nov/2025:11:53:43 +0530] "GET / HTTP/1.1" 200 468 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36 Edg/142.0.0.0"
192.168.1.9 - - [05/Nov/2025:11:54:14 +0530] "GET / HTTP/1.1" 200 468 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36 Edg/142.0.0.0"
192.168.1.9 - - [05/Nov/2025:11:54:37 +0530] "-" 408 0 "-" "-"
```

Step 8: Enable Firewall and Allow Web Traffic

```
sudo apt install ufw -y
sudo ufw enable
sudo ufw allow 80/tcp
sudo ufw status
```

Step 9: Block the Windows IP

```
sudo ufw deny from 192.168.1.9
```

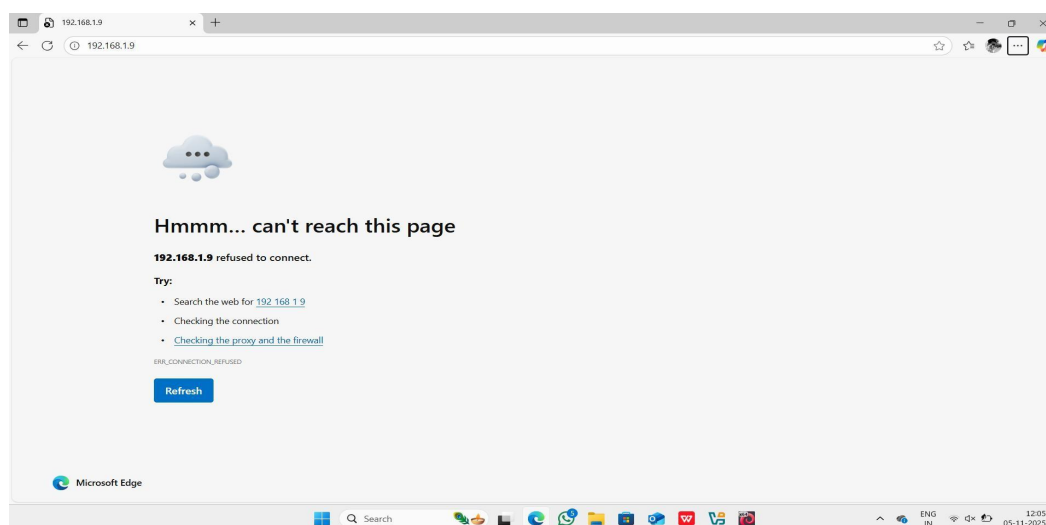
Check rules:

```
sudo ufw status numbered
```

```
root@kali:~/var/log/apache2#  
root@kali:~/var/log/apache2# sudo apt install ufw -y  
ufw is already the newest version (0.36.2-9).  
Summary:  
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 899  
  
root@kali:~/var/log/apache2#  
root@kali:~/var/log/apache2# sudo ufw enable  
Firewall is active and enabled on system startup  
  
root@kali:~/var/log/apache2#  
root@kali:~/var/log/apache2# sudo ufw allow 80/tcp  
Skipping adding existing rule  
Skipping adding existing rule (v6)  
  
root@kali:~/var/log/apache2#  
root@kali:~/var/log/apache2# sudo ufw status  
Status: active  
  
To Action From  
--  
80/tcp ALLOW Anywhere  
80/tcp (v6) ALLOW Anywhere (v6)  
  
root@kali:~/var/log/apache2#  
root@kali:~/var/log/apache2# sudo ufw deny from 192.168.1.9  
Rule added
```

Step 10: Verify Block Worked or not.

On Windows → refresh browser. The webpage should not load now.



❖ **Conclusion:**

In this project, we learned how to connect Windows and Kali Linux and make them communicate. We set up a web server on Kali and accessed the web page from Windows using the IP address. Finally, we saw how to check logs and block the Windows PC using a firewall for security.